ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

ROBOTICS AND INDUSTRIAL AUTOMATION

PAPER 1 · PRACTICE B

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Paper 1 style (theory). Section A: objective questions (30) · Section B: coordinate frames and kinematics, joint, linear and circular moves, robot programs and cycle time, machine vision and sensors in automation, types of automation, robot specifications and safety (70).

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INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
6. A non-programmable calculator may be used.
7. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The coordinate frame fixed to the mounting point of the robot is the …

A tool frame. B base frame. C user frame. D world frame.

1.1.2 The move in which the tool centre point travels in a straight line is a …

A joint move. B linear move. C circular move. D random move.

1.1.3 TCP stands for …

A tool centre point. B total control program.
C transfer control protocol. D tool cooling pump.

1.1.4 The type of automation used for very large volumes of one product is …

A flexible automation. B programmable automation.
C fixed (hard) automation. D manual production.

1.1.5 The sensor that lets a robot "see" the position of parts is a …

A machine-vision camera. B thermocouple.
C reed switch. D load cell.

1.1.6 The repeatability of a robot is given as, for example, …

A 20 kg. B 1,5 m. C 6 axes. D ±0,02 mm.

1.1.7 The international standard for the safety of industrial robots is …

A ISO 9001. B ISO 10218. C SANS 10142. D ISO 14001.

1.1.8 Inverse kinematics calculates …

A the joint angles needed to reach a tool position. B the tool position from the joint angles.
C the motor current. D the cycle time.

1.1.9 A force/torque sensor in the robot's wrist is most useful for …

A measuring colour.
B reading barcodes.
C counting parts.
D assembly that needs a controlled contact force.

1.1.10 To work safely beside people, a cobot mainly limits its …

A colour.
B paint.
C power and force.
D supply voltage only.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 Forward kinematics finds the tool position from the joint angles.

1.2.2 A joint (point-to-point) move always follows a straight line.

1.2.3 A user frame set up on a work table lets a program be moved easily to a new table position.

1.2.4 Flexible automation can change between different products with little changeover time.

1.2.5 Machine vision can check both the position and the quality of parts.

1.2.6 Cycle time is the time taken for one complete operation.

1.2.7 A safety PLC is simply a normal PLC with no extra checking.

1.2.8 Speed-and-separation monitoring slows a robot down when a person comes closer.

1.2.9 A robot's payload rating does not need to include the mass of the gripper.

1.2.10 Fixed automation is best for products that change every week.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Coordinate frame at the tip of the tool	A	reach
1.3.2	Move along an arc defined by three points	B	cobot
1.3.3	Working out the joint angles for a wanted tool position	C	tool frame
1.3.4	Time to complete one full operation	D	offline programming
1.3.5	Camera system that inspects and locates parts	E	cycle time
1.3.6	Robot designed to work safely beside people	F	waypoint
1.3.7	Recorded position that the robot moves to	G	circular move
1.3.8	Writing and testing a program on a computer model	H	programmable automation
1.3.9	Greatest distance the wrist can reach from the base	I	inverse kinematics
1.3.10	Automation reprogrammed between batches of products	J	machine vision

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: COORDINATE FRAMES AND KINEMATICS

2.1 Name and explain FOUR coordinate frames used when programming an industrial robot. (8)

2.2 Explain the difference between forward kinematics and inverse kinematics. (4)

2.3 A two-link planar arm has a first link of 0,4 m and a second link of 0,3 m. The first joint is at 30° to the X-axis and the second joint is at 0° (the arm is straight). Calculate the X and Y coordinates of the end of the arm. (4)

[16]

QUESTION 3: MOTION AND PROGRAMMING

3.1 Explain a joint move, a linear move and a circular move, and give ONE situation where EACH is used. (6)

3.2 Write the steps (pseudocode or robot instructions) of a pick-and-place program that uses approach points above the pick and place positions. (6)

3.3 One cycle takes 1,2 s to pick, 2,5 s to move, 1,0 s to place and 2,3 s to return. Calculate the cycle time and the number of parts handled per hour. (4)

3.4 Explain why approach points are used above pick and place positions. (2)

[18]

QUESTION 4: SENSORS AND VISION IN AUTOMATION

4.1 Explain how a machine-vision system works, from capturing the image to the decision. (4)

4.2 Give THREE applications of machine vision in a factory. (3)

4.3 Explain why a force/torque sensor is used in a robot wrist. (2)

4.4 Choose a suitable sensor for EACH task and give a reason: (a) detecting a cardboard box on a conveyor; (b) checking that a label is present and straight; (c) weighing a filled box; (d) measuring the angle of a robot joint. (8)

4.5 Why is good lighting important for machine vision? (1)

[18]

QUESTION 5: AUTOMATION, SPECIFICATIONS AND SAFETY

5.1 Compare fixed, programmable and flexible automation. For EACH, give ONE feature and ONE example. (6)

5.2 A robot has a payload rating of 10 kg. Its gripper has a mass of 2,5 kg and the part has a mass of 6 kg. Show whether the robot is suitable. (2)

5.3 Name FIVE measures that make an industrial robot cell safe. (5)

5.4 Explain speed-and-separation monitoring and power-and-force limiting, which allow cobots to work near people. (4)

5.5 What is the first step of a risk assessment for a new robot cell? (1)

[18]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER